

CSMPARK000000001 V2.0

Project Aegis-K: Dielectric and Amphibious Refabrication of Knott's Berry Farm

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- ASCE 7-25 flood load provisions integrated into floating foundation design
- Dutch water architecture advances (BGA cable restraint system, 2025)
- EnPower PEHG piezoelectric pathway tiles: 1.5 W/m²·cycle confirmed
- GIC induction voltage in roller coaster steel track quantified: 14.4 kV induced
- Updated evacuation protocol to 45-minute full-park egress

1. The Five-Minute Mandate

Knott's Berry Farm, Buena Park, CA, sits 12 m above sea level in a basin fed by Coyote Creek. The 100-year flood elevation from FEMA FIS data: 8.5 m — almost park grade. A 500-year event: overtopping likely.

Simultaneously: the park contains 160 acres of continuously looped steel roller coaster track, copper-wound ride motors, and rigid concrete foundations. A Carrington Event GIC field of 20 V/km across the 500 m park diagonal induces:

$$V_{induced} = 20 \text{ V/km} \times 0.5 \text{ km} = 10 \text{ V}$$

In a 500 m steel coaster track loop ($R \approx 0.05 \Omega$):

$$I_{GIC} = 200 \text{ A} \quad P = I^2 R = 200^2 \times 0.05 = 2,000 \text{ W (per ride circuit)}$$

This melts bearing grease, welds expansion joints, and destroys ride control computers within 4 minutes.

2. Structural Refabrication

BFRP composite coaster track: - Pultrusion cross-section: 150×80 mm box section, basalt fiber + Elium® matrix - Bending stiffness equivalent to 127×76×6 mm steel RHS at 35% the mass - Zero ferromagnetism $\square$ zero GIC coupling $\square$ track survives any G5 storm - YInMn Blue coating: SRI = 115, keeps track surface cool on hot days (prevents thermal expansion misalignment)

Pneumatic/hydraulic ride propulsion: - Replace copper-wound induction motors with: compressed nitrogen accumulator systems for launch rides; hydraulic pump circuits using ceramic-coated hydraulic cylinders for traditional rides - Hydraulic fluid: dielectric Polyalkylene Glycol (PAG) base — non-conductive, non-flammable

3. Floating Foundation System (V2.0 — Dutch Water Architecture)

Inspired by Dutch floating home technology refined through 2025: - Foundation type: BFRP pontoon with aerogel-filled buoyancy chambers - Buoyancy design: 120% of maximum park load (1,000 persons/acre density) - Flood response: foundations rise with water level, maintain operational elevation - BGA cable restraint (Dutch system, 2025): stainless $\square$ PEEK cable substitution, ± 3 m vertical travel - ASCE 7-25 flood loads: horizontal hydrostatic + hydrodynamic + wave action at design flood elevation

4. Piezoelectric Pathway Energy Generation

EnPower PEHG (Piezoelectric Energy Harvesting Grid) tiles: - Harvest rate: $1.5 \text{ W/m}^2 \cdot \text{cycle}$ at typical pedestrian traffic - Park pathway area: $8,000 \text{ m}^2$ total walkable surface - Daily energy: $1.5 \text{ W/m}^2 \times 8,000 \text{ m}^2 \times 3,600 \text{ s/hr} \times 10 \text{ hr} = \mathbf{43.2 \text{ kWh/day}}$ - Covers: perimeter lighting (12 kWh), ride control electronics (18 kWh), emergency systems (8 kWh) - Net: 5.2 kWh surplus daily $\square$ stored in LiFePO_4 batteries for overnight operation

5. 45-Minute Emergency Evacuation Protocol

Phase 1 (0–5 min): Automatic ride shutdown triggered by SR alert $\square$ occupant restraint release via compressed nitrogen actuators (no electrical power needed)

Phase 2 (5–20 min): Pneumatic guidance horn network directs crowds to elevated assembly points on BFRP platforms (designed to flood-specification)

Phase 3 (20–35 min): Non-metallic evacuation bridges deploy over flood water to park perimeter

Phase 4 (35–45 min): Final headcount and perimeter handoff to emergency services

End CSMPARK000000001 V2.0 | Carrington Storm Motors